Radar One-on-One

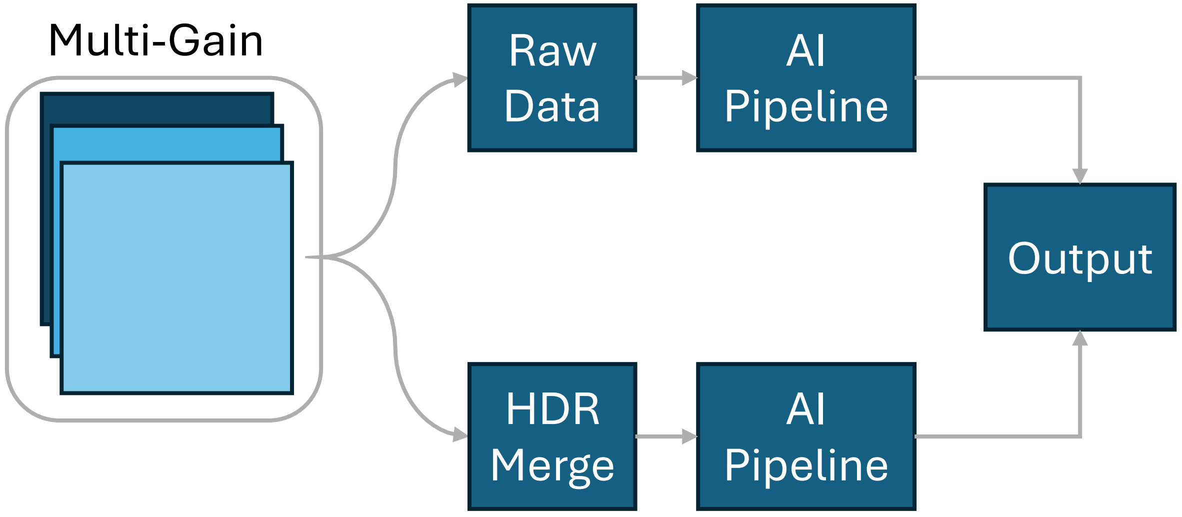
Aug 14, 2025

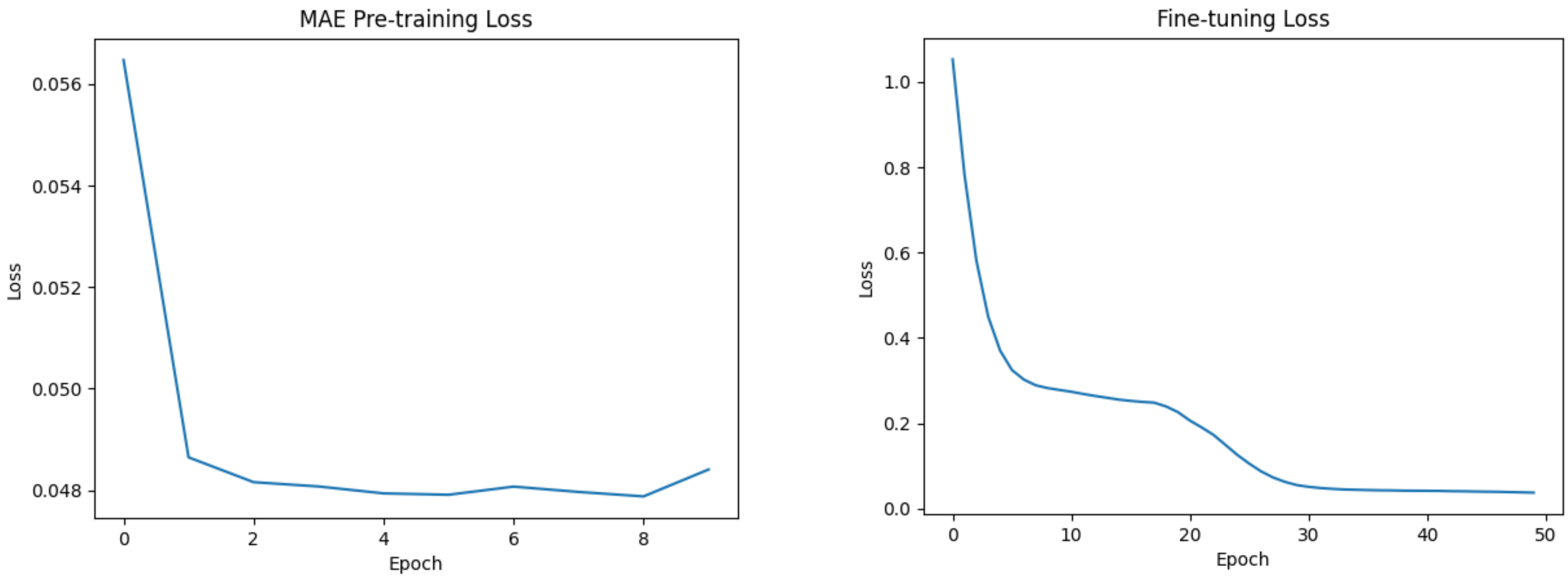
Object Detection

- Few-Shot learning
  - Human-labelled 3 images
  - Have a data set of about 6k images that are good for training.
  - Need to improve the data augmentation process. (i.e. rotate the images by some random amount and fill in blank space with black pixels)
  - Masked AutoEncoder -> Lightweight detection head(I need to look more into

Model types roughly

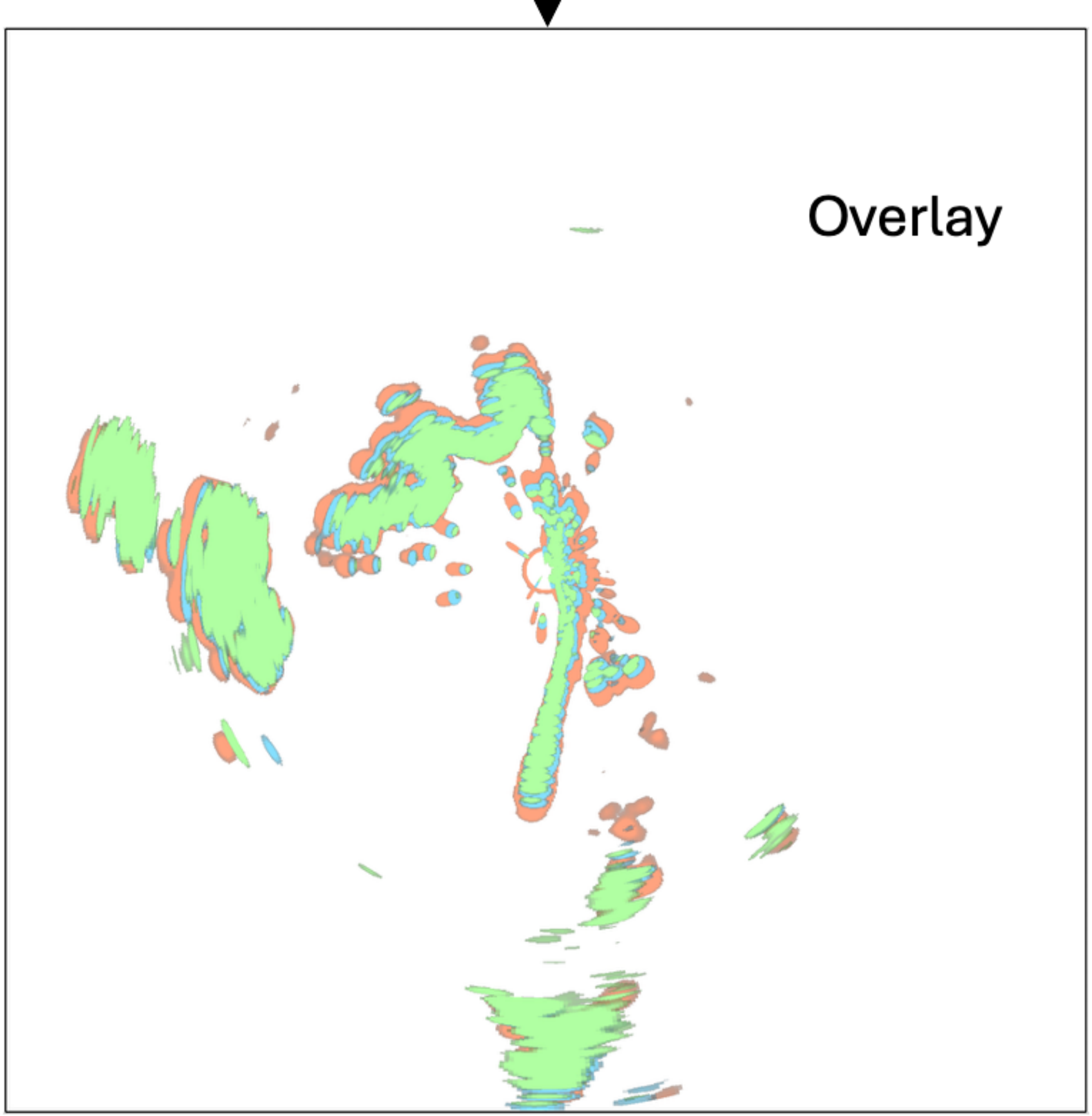
- Input: bursts of RA frames at gains 40/50/70; option to fuse into HDR or feed raw.
- No Doppler: speed from inter-frame motion (tracking or optical-flow)
- Few-Shot: ~3-15 labeled frames + thousands unlabeled → self-supervised pretrain + pseudo-labeling + imprinted prototypes.

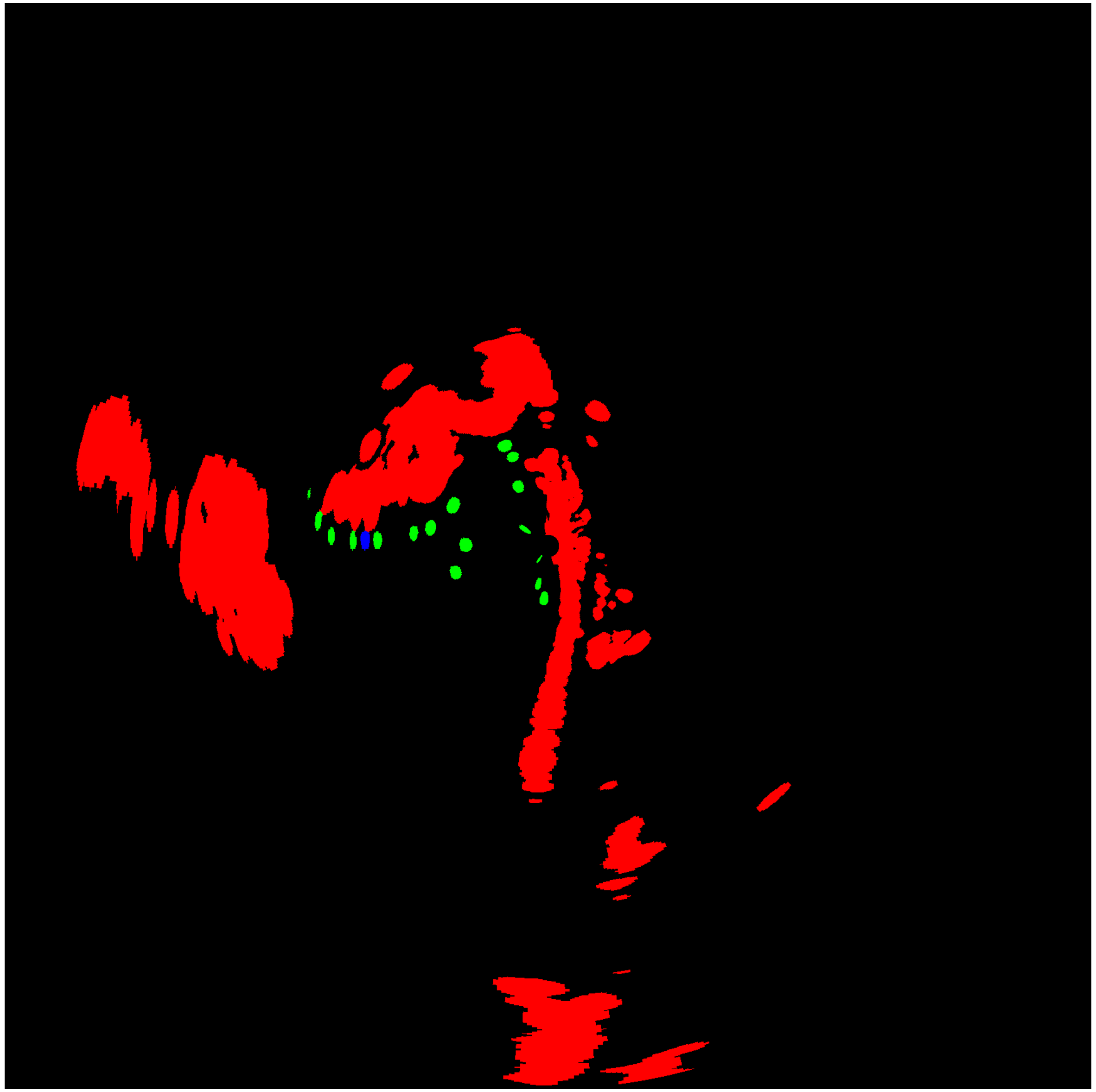


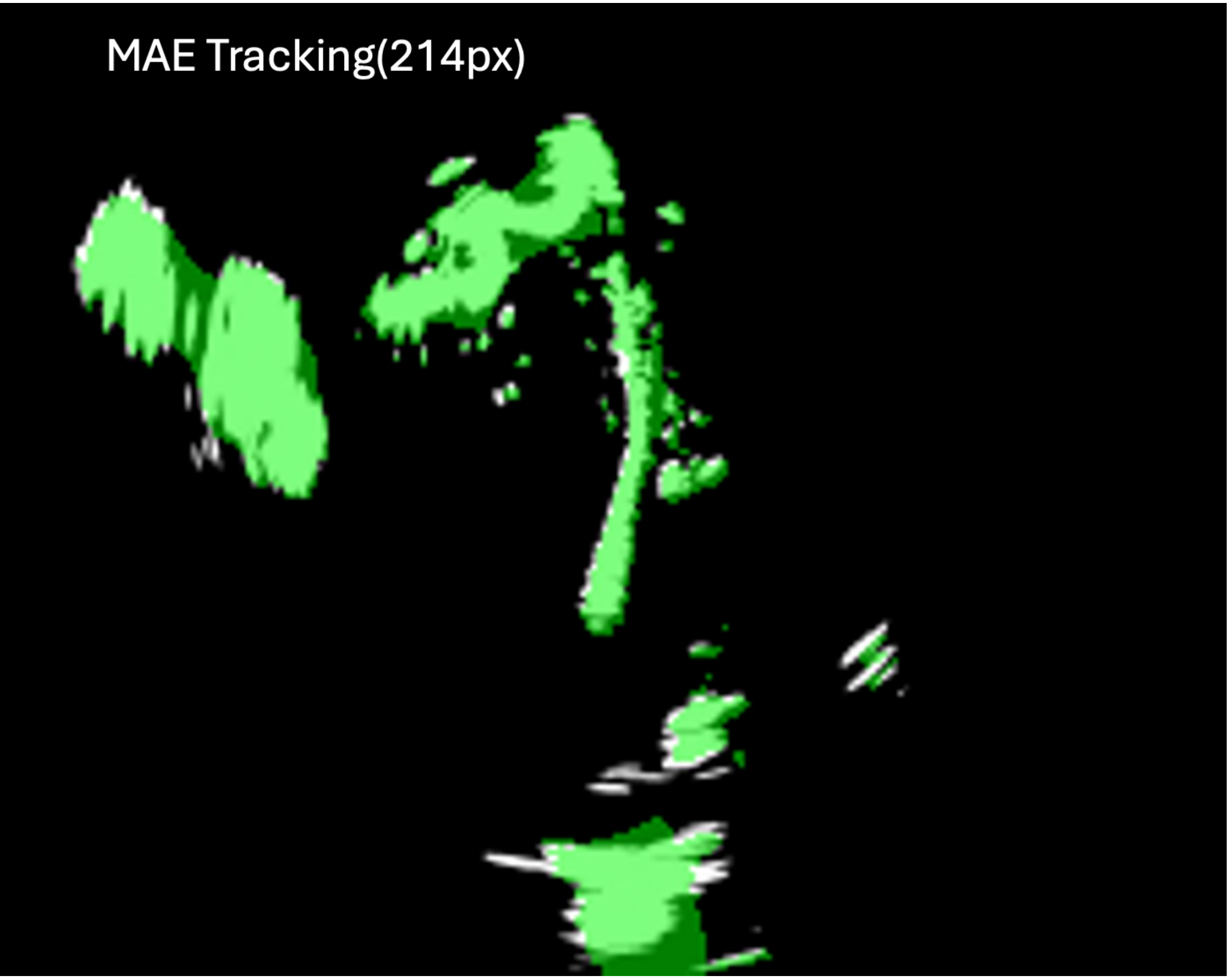

MAE Pre-training Loss
Loss
Epoch
0.056
0.054
0.052
0.050
0.048
0
2
4
6
8
Fine-tuning Loss
Loss
Epoch
1.0
0.8
0.6
0.4
0.2
0.0
0
10
20
30
40
50

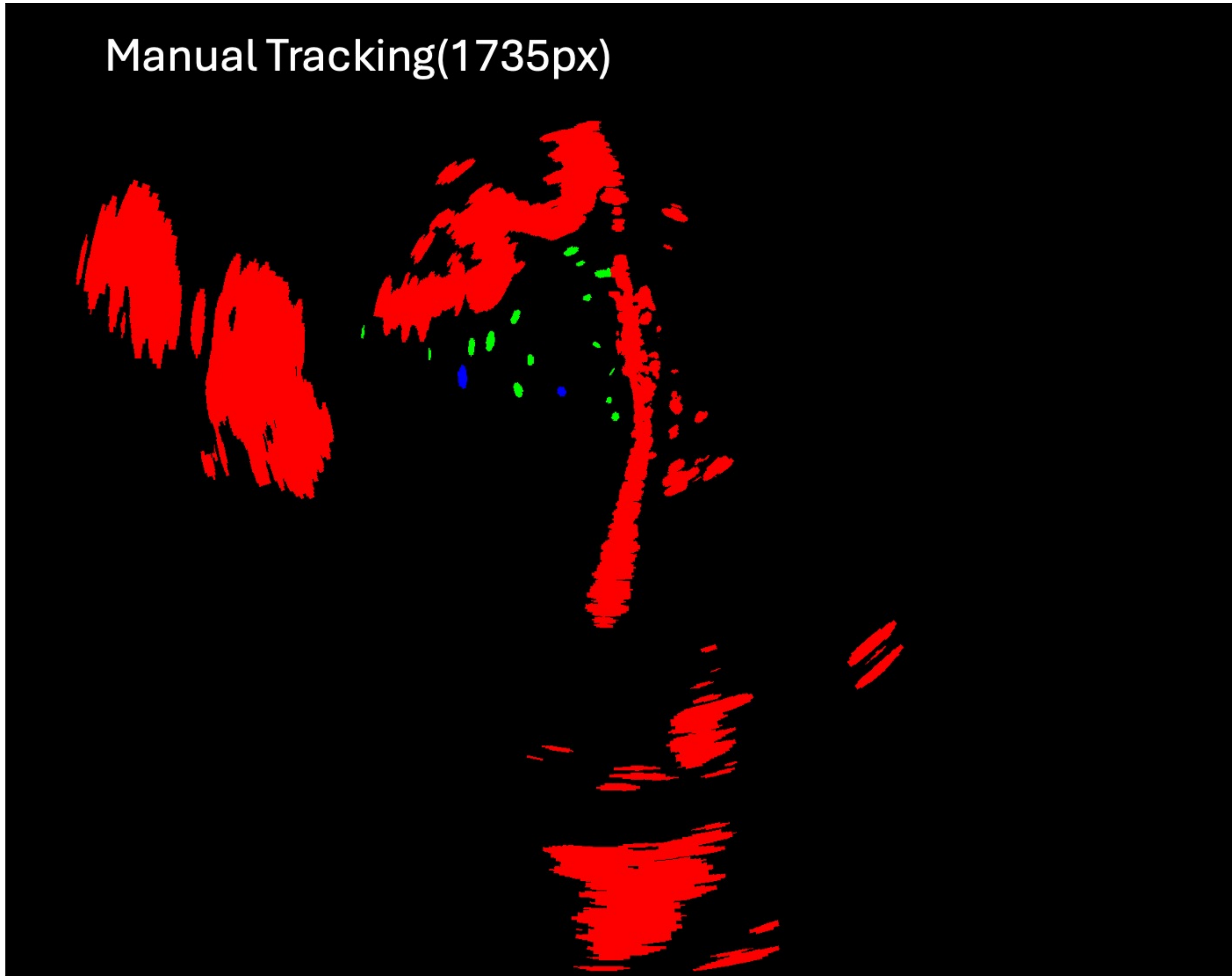
HDR Video

- For QPR ill show 40 gain v HDR – and a comparison of objects NOT detected bc of single Gain


Overlay




MAE Tracking(214px)


Manual Tracking(1735px)